

MonReader - CV Interview Questions

Part A is holistic CV and DL concepts. Part B is what interviewers ask if you put MonReader on your resume.

Part A: Holistic CV and DL Interview Questions

A1. Neural Network Fundamentals

1. Walk me through backpropagation in one paragraph.
2. What is the vanishing gradient problem? How do modern architectures fight it?
3. What is the exploding gradient problem? How do you detect and fix it?
4. Compare ReLU, LeakyReLU, ELU, GELU, and Swish. When would you use each?
5. Why is Sigmoid a bad activation for hidden layers in deep nets?
6. What is the universal approximation theorem, and why does it not mean neural nets solve everything?
7. Explain the bias-variance tradeoff in the context of deep learning.

A2. Convolutional Neural Networks

1. What is a convolution operation? How is it different from a fully connected layer?
2. Explain filters, strides, and padding. What does "same" vs "valid" padding do?
3. What is the receptive field? How does it grow with depth?
4. What is pooling and why do we use it? MaxPool vs AveragePool vs GlobalAveragePool.
5. Why do we stack multiple small 3x3 convs instead of one large 7x7 conv?
6. What is a 1x1 convolution and when is it useful?
7. Explain filters, strides, and padding. What does "same" vs "valid" padding do?

A3. Architecture Internals (Important for This Project)

1. Walk me through the VGG16 architecture. Why is it called "16"?
2. What is a residual connection in ResNet? Why does it help train deeper networks?
3. Explain the Inception module. Why parallel branches?
4. What is a depthwise separable convolution? How does it differ from a standard convolution?
5. Walk me through the math of why depthwise separable convolutions are cheaper.
6. What is an inverted residual and linear bottleneck (MobileNetV2)?
7. Explain EfficientNet's compound scaling. Why did they scale depth, width, and resolution together?
8. Compare VGG16, ResNet50, MobileNetV2, and EfficientNetB0 on params, FLOPs, and typical ImageNet top-1 accuracy.

A4. Regularization and Normalization

1. What is dropout? What happens at inference time?
2. What is batch normalization? Walk me through the exact operations.
3. Where does batch norm sit relative to the activation function?
4. What problem does batch norm actually solve? (Hint: the original paper says internal covariate shift, but recent papers argue otherwise.)
5. Compare batch norm, layer norm, instance norm, and group norm. When is each preferred?
6. Why does batch norm behave differently in train mode vs eval mode?
7. What is weight decay? How is it different from L2 regularization in Adam?

A5. Data Augmentation

1. Why do we use data augmentation? What is the intuition?
2. List 10 augmentation techniques for images.
3. What is Mixup? What is CutMix? How do they differ?
4. Should you augment the validation and test sets? Why or why not?
5. What is Test-Time Augmentation and when would you use it?

A6. Transfer Learning

1. Explain feature extraction vs fine-tuning.
2. When should you freeze all base layers vs unfreeze only the last few vs unfreeze everything?
3. Why do we use a very small learning rate when fine-tuning?
4. When would transfer learning fail? (Domain gap, tiny target dataset with tiny models, etc.)
5. Why is ImageNet pretraining useful for medical or satellite images even though the domains are different?

A7. Loss Functions

1. Why is cross-entropy the standard loss for classification?
2. Derive binary cross-entropy from likelihood.
3. What is focal loss? When would you use it over cross-entropy?
4. What is label smoothing and why does it help?
5. What is Dice loss? When is it preferred? (Segmentation context)

A8. Training Dynamics and Optimizers

1. Compare SGD, SGD with momentum, RMSprop, Adam, and AdamW.
2. What is a learning rate schedule? Compare step, cosine, and warm-up schedules.

3. What is the "one cycle" policy?
4. What are the signs that your learning rate is too high or too low?
5. What does gradient clipping do?

A9. Evaluation Metrics

1. When is accuracy a bad metric?
2. Explain precision, recall, and F1. When do you optimize for each?
3. What is ROC-AUC? What does 0.5 mean? What is its limitation?
4. Compare ROC-AUC and PR-AUC. When is PR-AUC preferred?
5. How would you evaluate a model on an imbalanced dataset?

A10. Interpretability

1. What is Grad-CAM? Walk me through the algorithm.
2. What is the difference between Grad-CAM and saliency maps?
3. What are SHAP and LIME? Are they used in CV?
4. Why is interpretability important in a production CV system?

A11. Model Deployment and Quantization

1. What is quantization? Compare int8, float16, and dynamic quantization.
2. What is post-training quantization vs quantization-aware training?
3. What is model pruning? Compare structured and unstructured pruning.
4. What is knowledge distillation?
5. What is TensorFlow Lite? What is ONNX? Why do they exist?
6. Why does model size matter for mobile deployment beyond just download size?

A12. Breadth (Should Know Exist)

1. What is a Vision Transformer? How is it different from a CNN?
 2. What is U-Net used for?
 3. What is YOLO in one sentence?
 4. What is a GAN? Generator and discriminator, high level.
 5. What is a diffusion model, at a high level?
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Part B: MonReader Project-Specific Questions

B1. Problem Framing

1. Walk me through this project in 2 minutes.
2. Why is this a binary classification problem and not multiclass?

3. The problem allows sequence-based prediction as a stretch. Why did you stick to single-image?
4. What were the two main constraints on this project, and why did they matter?

B2. Data and Preprocessing

1. Walk me through your EDA. What did you find?
2. Was your dataset class-balanced? How did you check, and what would you have done if it were not?
3. What image size did you settle on and why?
4. Why did you use each model's own preprocess_input instead of just dividing pixels by 255?
5. How did you split training into train and validation? Why that split?
6. What data augmentations did you use? Why those specifically?
7. Why is augmentation only applied during training and not on validation or test?

B3. Model Selection and Architecture

1. Why did you try 6 different models instead of just picking one?
2. Walk me through your custom CNN. Why those filter sizes, depths, and activations?
3. What did you learn from training a CNN from scratch vs using transfer learning?
4. Walk me through the difference between VGG16 + FF and VGG16 + Augmentation. What did the numbers tell you?
5. Why is ResNet50 easier to train than VGG16 despite being deeper?
6. Why does MobileNetV2 have so many fewer parameters than ResNet50?
7. What are depthwise separable convolutions? Explain them to a non-ML engineer.
8. Why did you pick MobileNetV2 as your final model?
9. What would happen if you tried MobileNetV3 or EfficientNet-Lite? Would they be better?

B4. Training Setup

1. What optimizer, learning rate, and batch size did you use? Why?
2. Did you use any callbacks? Which ones and why?
3. What is your fine-tuning strategy? Walk me through freezing vs unfreezing.
4. Why did you use a lower learning rate during fine-tuning?
5. How did you decide when to stop training?

B5. Evaluation

1. Why F1 and not accuracy?
2. Walk me through your comparison table. What is the story it tells?
3. Which metric surprised you the most?
4. If your F1 on training was 0.99 but on test was 0.85, what would you do?
5. What is your confusion matrix telling you? Where does the model fail most?

6. Why is ROC-AUC not enough on its own?

B6. Interpretability

1. Walk me through your Grad-CAM setup.
2. What did Grad-CAM show you? Was the model looking at the right regions?
3. Did Grad-CAM reveal any shortcut learning? (For example, backgrounds instead of hands or pages.)
4. How would you use Grad-CAM insights to improve the model?

B7. Size Constraint and Deployment

1. Why does the 40 MB budget matter?
2. Which of your models fit under 40 MB? Which did not?
3. If MobileNetV2 was still too large, what would you try next?
4. What is quantization? How much size reduction did you get (if you tried it)?
5. What is pruning? How is it different from quantization?
6. What is knowledge distillation? Would it help here?
7. Why can't we just use ResNet50 and compress it heavily?

B8. App and Productionization

1. Walk me through your Gradio app architecture.
2. What happens when a user uploads an image?
3. How would you turn this into a real-time video pipeline?
4. If the model was deployed on-device, what would the memory and latency profile look like?
5. How would you monitor this model in production?

B9. Failure Modes and Edge Cases

1. What kinds of images does your model get wrong?
2. What would happen if the user's camera has low light or motion blur?
3. What if the user is not flipping a book but a magazine or a laptop?
4. How would you handle out-of-distribution inputs?

B10. Stretch and Curveballs

1. If you had labels for individual video frames as sequences, how would your approach change? (Answer should mention 3D CNNs, LSTMs, or transformer-based video models.)
2. What is the difference between action recognition and image classification? Which is closer to what MonReader actually needs in production?
3. If given a month more, what would you do next?

4. What is the hardest part of this project?
 5. What would you do differently if you did this project again?
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How to Use These Questions

- After each week of the learning path, come back to Part A and answer the relevant questions out loud. If you cannot say it clearly, you do not know it yet.
 - Before adding this project to your resume, do a full mock walkthrough using Part B. With mentor, or record yourself. Whichever works.
 - Keep growing your weak-areas doc from the NLP project. Add MonReader gaps there too.
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